



QUALITY SYSTEM

FORM

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TITLE:

LABORATORY OOS INVESTIGATION FORM

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OOS Number: OOS- 262080

Phase I

Section A – Test Details

Initiator:	America Alanis (Written by Qiyue Chen)	Test Date:	01-Sep-2026	Date Initiated:	08-Sep-2026	
Date of Incident:	08-Sep-2026	Test Name:	Celsis Sterility	Sample ID:	ETX-260828-0527	
Name of Analyst who Performed the Test:	Andrew Carrillo (AC) (Celsis Sterility Prepper) Gabrielle Surber (GS) (Celsis Sterility Processor) Cuong Du (CCD) (Celsis Sterility Aliquoting Analyst) America Alanis (ALA) (Celsis Sterility Reading Analyst)	Sample / Active Name	MOTs-C 10 MG/ML (5 ML) Injection	Lot #:	LG342010349	
		Dosage Form	Injection			
Description of Incident:	On 08 Sep 2026, sample ETX-260828-0527 was found to be positive for growth after Celsis Sterility Testing ☒ Client Care notified of OOS ☒ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS					
	SOP / Test Method # MICRO-SOP-44	Effective Date: 03-Jan-2025	SOP / Test Method Rev: 01	Limits/Specification: Pass/Fail		
Manager notified	Manager/designee name: Kathan Parikh	Date: 08-Sep-2026	Quality Assurance Department notified:	Yes ☒ No ☐		

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy			Comments	
Analyst interviewed?	☒ Yes ☐ No ☐ N/A		Yes, analysts Andrew Carrillo, Gabrielle Surber, America Alanis & Cuong Du were comprehensively interviewed.	
Correct samples analyzed?	☒ Yes ☐ No ☐ N/A		Yes, Sample ID: ETX-260828-0527	
Correct SOP / Test Method used/followed?	☒ Yes ☐ No ☐ N/A		Yes, as per MICRO-SOP-44	
Correct sample collection/technique?	☒ Yes ☐ No ☐ N/A		Yes, as per MICRO-SOP-44	
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☐ No ☐ N/A		Yes, Information is available in Eagle	
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A		Yes, all analysts are trained and	
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A		Not Applicable	
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☒ N/A		Not Applicable	
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A		Yes, the samples were stored	
Material Discrepancy			Lot Number	Expiry Date
Is the integrity of the sample(s) questionable?	☐ Yes ☒ No ☐ N/A			
Was the sample transported appropriately?	☒ Yes ☐ No ☒ N/A			
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A			



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Appropriate/correct glassware used?	☐ Yes ☐ No ☒ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Materials and Reagents TSB: 06242026-1 FTM: 06162026-2 PBS (Phosphate Buffered Saline): 35725002	Materials and Reagents TSB: 22 Sep 2026 FTM: 14 Sep 2026 PBS: 31 Dec 2027 Bovine Serum Albumin (BSA): 23 Jan 2028
Standard Used	Celsis ATP Positive Control	022601-1483	31 Jan 2027
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☒ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☒ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A	E002222	Sep 2026
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A	E001736 (114),	Dec 2026, Jun
System Performance Check (if applicable)?	☒ Yes ☐ No ☐ N/A	E002222	Sep 2026
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☒ Yes ☐ No ☐ N/A	See Phase I	See Phase I
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☒ Yes ☐ No ☐ N/A	See Phase I	See Phase I
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☐ N/A	Incubator E001356 (Monitored by Sensor E001450) Incubator E001357 (Monitored by Sensor F001449)	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027

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Are chromatogram/spectra/raw data correct?				☐ Yes	☐ No	☒ N/A	Not Applicable	Not Applicable
Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)								
☒ N/A		☐ Re-vialing of original working sample solution						
☐ Re-injection of sample vial		☐ Re-dilution of a working sample solution from the original stock						
☐ Re-injection of original standard vial		☐ Re-dilution of a working standard solution from the original stock standard						
☐ Re-vialing of original working standard solution		☐ Other: N/A QYC 08Sep2026						
Section D – Phase I Summary								
Phase I Summary: All analysts involved in the prepping, processing, aliquoting, and reading of the samples – Andrew Carrillo (AC), Gabrielle Surber (GS), America Alanis (ALA) and Cuong Du (CCD) were interviewed comprehensively. Their answers are recorded throughout this document. Upon arrival, the sample was stored in accordance with the Client's instructions. Both the sample prepping analyst - Andrew Carrillo (AC) - and the processing analyst - Gabrielle Surber (GS) - inspected the sample during preparation and processing. No leaks or visible turbidity were observed during preparation or processing; no evidence of compromised container integrity was identified during these observations. All reagents and supplies mentioned in the material section above were stored according to the suppliers' recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system. The sample was processed within the ISO 5 Biological Safety Cabinet (BSC), E001316, situated in the innermost cleanroom (114B), which connects to the intermediate ISO 7 buffer room (114A). This buffer room, in turn, leads to the outermost ISO 8 Anteroom (114). Furthermore, a positive air pressure system is maintained, ensuring airflow from the innermost ISO 7 cleanroom to the intermediate ISO 7 buffer room and finally to the outermost ISO 8 Anteroom. The innermost ISO 7 cleanroom 114B and its ISO 5 BSC E001316 were thoroughly cleaned and prepared before initiating the testing by analyst Gabrielle Surber as per MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility), and MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Similarly, for Celsis aliquoting, the ISO 7 cleanroom 114A and its ISO 5 BSC E001798 were thoroughly cleaned and prepared by analyst Cuong Du before initiating the testing as per MICRO-SOP-44 and MICRO-SOP-9. Both the BSCs, E001316 in 114B and E001798 in 114A, were certified and approved by the Engineering and Quality Assurance teams prior to use.								

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Phase I Summary (continued):

On 01 Sep 2026, the sample vials for ETX-260828-0527 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each sample vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. The secondary disinfection happened in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach with a 10-minute contact time before being transferred to the innermost ISO 7 cleanroom (114B). Inside 114B, the processing analyst, Gabrielle Surber, performed a final disinfection step with a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316. Inside the BSC, the vials were placed on the disinfected work surface, aseptically opened, and tested in accordance with MICRO-SOP-44 (Celsis Sterility Testing). Direct inoculation was performed by adding 25 mL per media (5 x 300mL FTM jars and 5 x 300mL TSB jars; 1g of Bovine Serum Albumin in 100mL of PBS and 5mL added per suitability method 2601120497). Following testing, the media bottles were subsequently transferred into designated incubators, E001356 and E001357, to initiate incubation.

Upon completion of incubation on 08 Sep 2026, TSB & FTM bottles for ETX-260828-0527 were disinfected and transferred to the intermediate ISO 7 buffer room (114A) for the aliquoting step per MICRO-SOP-44 (Celsis Sterility Testing). In 114A, the media bottles were disinfected one more time before transferring them to the ISO 5 BSC E001798 located in cleanroom 114A. In ISO 5 BSC E001798, the sample was aliquoted into assay cuvettes by analyst Cuong Du (CCD). After aliquoting, Celsis Sterility Reading was performed in accordance with MICRO-SOP-44 on Celsis instrument E002222 by analyst America Alanis (ALA).

Celsis testing produced a positive result for FTM sample container ETX-260828-0527-3/5 in accordance with the predefined decision criteria specified in MICRO-SOP-44. The FTM sample yielded 7190 RLU, which exceeded the method cutoff of 2955 RLU (FTM negative control = 985 RLU). The corresponding TSB sample container (ETX-260828-0527-4/5) tested negative with 1500 RLU (TSB cutoff = 2007 RLU, negative control = 669 RLU). Duplicate readings met the method-defined %CV criterion of < 30%. Furthermore, the Daily Controls—including the Instrument Blank, Reagent Blank, and ATP Positive Control (Positive Cut-off RLU = 102383)—met their respective predefined acceptance criteria with %CV below 30% as documented in the associated Celsis data packet.

Following the OOS result, the positive FTM bottle for ETX-260828-0527 was submitted for Differential Staining and Microbial Identification under ETX-260908-0593. No microbial growth was recovered by subculture (0 CFU on plate; see Table 1), and no cellular morphology was visualized by differential staining. Therefore, no organism was recovered for identification or further microbiological comparison. The lack of recovery on subculture does not invalidate the initial analytical Celsis result and may be consistent with low-level, stressed, or otherwise non-recoverable microbial cells under the test conditions; however, a specific microbial physiological state, including a viable but non-culturable (VBNC) state, could not be confirmed.

The culture media utilized were within their expiry period. The negative culture media bottles for the direct inoculation method for the original culture were handled, processed, and incubated in a manner identical to that of actual samples. No microbial growth was observed in the corresponding negative control.

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 Biosafety Cabinets for Cleanroom 114 was performed on 30 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). During cleaning, it was documented that all H2O2 indicators passed and met the applicable acceptance criteria.

Tables 2 & 3 (please see attached) present the environmental monitoring results for the duration of testing. The environmental monitoring (EM) plates were incubated for no less than 48 hours at 30-35°C and no less than an additional five days at 20-25°C as per MICRO-SOP-2 (Environmental Monitoring of the Cleanroom Facility). Table 2 pertains to Environmental Monitoring performed during Celsis Sterility Processing, and Table 3 pertains to Environmental Monitoring performed during Celsis Sterility Aliquoting.

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Phase I Summary (continued):

Environmental Monitoring from Celsis Sterility Processing:

A review of the Environmental Monitoring (EM) results for the relevant testing period showed no microbial growth on personnel monitoring plates, settling plates, or ISO 5 BSC (E001316) surface sampling plates on the date of testing (01Sep2026), the preceding sampling date (31Aug2026), or the subsequent sampling date (02Sep2026).

Environmental Monitoring from Celsis Sterility Aliquoting:

After reviewing the Environmental Monitoring (EM) results for the relevant aliquoting and reading dates, no microbial growth was observed on personnel monitoring plates for analyst CCD on the date of testing (08Sep2026), the preceding sampling date (03Sep2026), or the subsequent sampling date (15Sep2026). Similarly, ISO 5 BSC (E001798) surface sampling and settling plates demonstrated no microbial growth on the date of testing (08Sep2026, CCD), the preceding sampling date (07Sep2026, ALA), or the subsequent sampling date (09Sep2026, ALA).

Personnel monitoring, ISO 5 BSC surface monitoring, and ISO 5 settling plates collected on the respective processing (01Sep2026) and aliquoting (08Sep2026) dates showed no microbial recovery (0 CFU). Weekly active-air monitoring performed for the corresponding weeks identified 1 CFU in the outermost ISO 8 anteroom (114) on 04Sep2026 (under ETX-260914-0487, characterized as Gram (+) coccobacilli and Gram (+) rods) and 1 CFU in the same classified area on 10Sep2026 (under ETX-260921-0520, identification currently in progress); no microbial recovery was observed from the ISO 7 cleanrooms (114A and 114B) during either weekly active-air monitoring event. Weekly surface sampling across Suite 114 on both dates likewise demonstrated no microbial recovery across all evaluated locations. These data did not identify microbial recovery within the monitored ISO 5 critical processing areas or from the personnel monitoring associated with the testing activities.

All sample processing and aliquoting activities were performed within validated ISO 5 Biological Safety Cabinets (BSCs E001316 and E001798), located in ISO 7 cleanrooms 114B and 114A. The sample containers and media remained enclosed and protected during transfer through the ISO 8 anteroom and were not directly exposed to ISO 8 room air. Because no organism was recovered from ETX-260828-0527 during subsequent subculture, a direct microbiological comparison between the OOS sample and the ISO 8 environmental recoveries could not be performed. The ISO 8 findings therefore cannot be microbiologically linked to, or definitively excluded as related to, the OOS result on the basis of organism identification. Nevertheless, review of the enclosed material-transfer process, same-day ISO 5 and personnel monitoring (0 CFU), and facility airflow segregation records did not identify evidence linking the OOS result to the monitored laboratory environment.

All interviewed analysts reported following cleaning and handling procedures in accordance with MICRO-SOP-9 and MICRO-SOP-44. A review of the available records and facility monitoring data did not identify documented procedural deviations or abnormal equipment parameters during the processing and aliquoting activities.

ETX-260828-0527 was the first sample processed by analyst GS on 01 Sep 2026. A review of the remaining samples processed by GS that day and of the other samples processed, incubated, aliquoted, and read within the associated testing batch identified no additional positive Celsis results. Accordingly, the reviewed batch data did not identify a pattern supporting sample-to-sample cross-contamination during processing or aliquoting.

An analysis of the six-month sample history for Optimal Balance Pharmacy (E19193) indicates that Eagle Analytical processed samples for Celsis Sterility testing with no prior occurrences of an out-of-specification or positive result for this analyte during this period.

A review of the lot history shows that there have been no additional submissions of sample lot: LG342010349 for retesting using Celsis Sterility Testing or any other sterility method.

Based on the laboratory investigation, no assignable laboratory cause was identified involving the analyst, instrument, reagents, supplies, or monitored laboratory environment. The initial Celsis OOS result therefore remains valid in accordance with the applicable laboratory OOS procedure. Subsequent microbial identification testing did not recover an organism; consequently, the organism identity and source of the positive result could not be determined within the laboratory investigation. Further investigation and disposition, if required, should be performed in accordance with the applicable OOS procedure and client quality requirements.



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Has root cause been identified as a possible laboratory error?*Note: Root cause must be conclusive, verifiable, and well-documented.*☐ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☒ No, the OOS result is valid.**Reviewer comment(s):****Prepared by (Print):**

Qiyue Chen

Signature/Date:

SIGN

Lab Manager/Designee (Print):

Robin Seymour

Signature/Date:

SIGN

Section E – Quality Disposition**Can investigation be closed?**☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.**Approved by QA Representative (Print):**

Signature/Date:

SIGN

Table 1: Information for ETX-260828-0527 under investigation

Processing Analyst	Aliquoting Analyst	Sample ID	Related Microbial ID	Media with microbial growth	Microbial ID
Gabrielle Surber	Cuong Du	ETX-260828-0527	ETX-260908-0593	1 x 300mL FTM	No growth was obtained

Table 2: Environmental Monitoring from Processing Performed on 01Sep26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	31Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	01Sep26	GS	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	02Sep26	GS	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC)								
Surface Sampling of ISO 5 (4 locations)	Daily	31Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	01Sep26	GS	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	02Sep26	GS	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	31Aug26	GS	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	01Sep26	GS	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	02Sep26	GS	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling Bracketing								
Active Air Sampling of Cleanrooms	Weekly	04Sep26	ISS	Week on Testing Date	1 CFU (ISO 8 114)	ETX-260914-0487	Gram (+) coccobacilli & Gram (+) rods	None
Surface Sampling of Anteroom and Cleanroom Bracketing								
Surface Sampling of Cleanrooms	Weekly	04Sep26	ISS	Week on Testing Date	No growth	N/A	N/A	None

Table 3: Environmental Monitoring from Aliquoting Performed on 08Sep26

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMMYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personnel (Left Touch and Right Touch)	Daily	03Sep26	CCD	Date Before Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	08Sep26	CCD	Date of Testing	No growth	N/A	N/A	None
Personnel (Left Touch and Right Touch)	Daily	15Sep26	CCD	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC)								
Surface Sampling of ISO 5 (4 locations)	Daily	07Sep26	ALA	Date Before Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	08Sep26	CCD	Date of Testing	No growth	N/A	N/A	None
Surface Sampling of ISO 5 (4 locations)	Daily	09Sep26	ALA	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	07Sep26	ALA	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	08Sep26	CCD	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	09Sep26	ALA	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling Bracketing								
Active Air Sampling of Cleanrooms	Weekly	10Sep26	SMO	Week on Testing Date	1 CFU (ISO 8 114)	ETX-260921-0520	Pending	None
Surface Sampling of Anteroom and Cleanroom Bracketing								
Surface Sampling of Cleanrooms	Weekly	10Sep26	SMO	Week on Testing Date	No growth	N/A	N/A	None